Inorganic Cumulative Examination
October 19, 2007

Hydrogen has a number of unique characteristics and it is important in a number of areas of chemistry. Two quotes from Greenwood and Earshaw illustrate the point. " In fact hydrogen forms more compounds than any other element including carbon and a survey of its chemistry, therefore, encompasses virtually the entire periodic table." "Despite its very simple electronic configuration hydrogen can, paradoxically, exist in over 50 different forms, most of which have been well characterized."

1. The properties of water are very much influenced by the presence of hydrogen and the associated hydrogen bond. Define the meaning of hydrogen bond. Describe the physical properties of water in the absence of hydrogen bonding.

2. It has been suggested that hydrogen could be placed in Group 1, Group 14 or Group 17 on the periodic table. Provide arguments based on specific chemical reactions with specific compounds (no generalizations) for each of these suggestions.

3. One of the most important industrial applications of hydrogen is the production of ammonium nitrate. Describe the chemical reactions involved in the preparation of ammonium nitrate on an industrial scale. Do not forget the reactions that provide the hydrogen gas.

4. A "hydrogen economy" is based on an inexpensive source of hydrogen that does not rely on the use of fossil fuels. Describe in detail a chemical scheme that satisfies the criteria.

5. What are ortho and para hydrogen?

6. What is "cold fusion"? Be very specific.

7. What is tritium?

8. Describe the preparation of diborane.